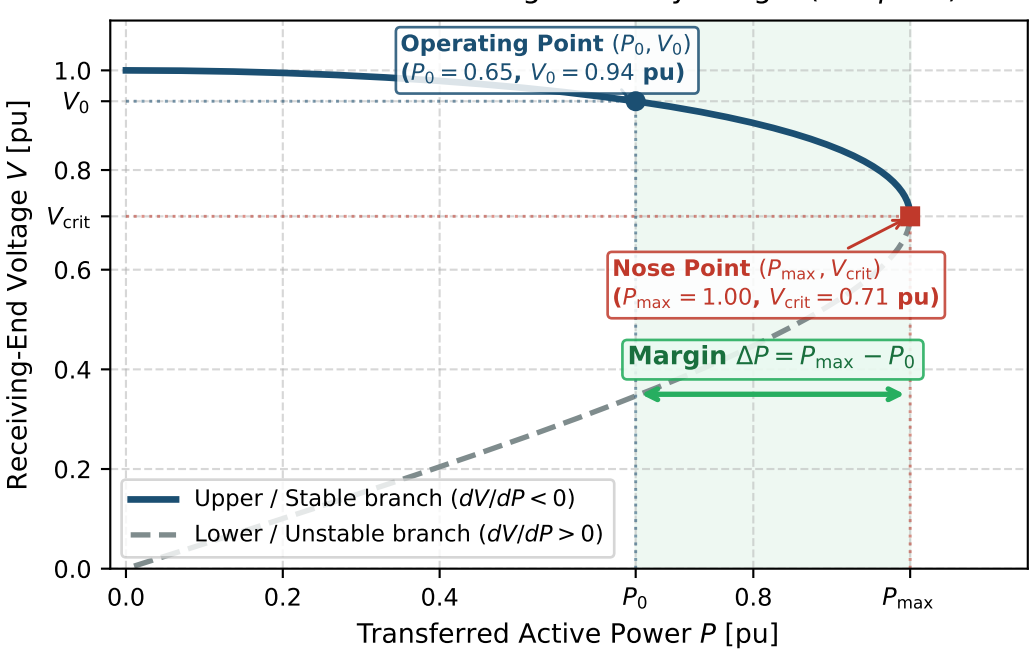
P-V Nose Curve & Voltage Stability Margin (cos ϕ = 1)
Operating Point (P0, V0)
(P0 = 0.65, V0 = 0.94 pu)
Nose Point (Pmax, Vcrit)
(Pmax = 1.00, Vcrit = 0.71 pu)
Margin ΔP = Pmax − P0
Upper / Stable branch (dV/dP < 0)
Lower / Unstable branch (dV/dP > 0)
Receiving-End Voltage V [pu]
Transferred Active Power P [pu]
1.0
V0
0.8
Vcrit
0.6
0.4
0.2
0.0
0.0
0.2
0.4
P0
0.8
Pmax